

Introduction to GitHub

What is GitHub?

GitHub is a website used to store coding projects online. It helps students save their work safely and access it anytime from different computers.

Many programmers and students use GitHub to:

- Save coding files online
- Share projects with friends or teammates
- Work on projects from school or home
- Keep track of changes made in a project

You can think of GitHub as an online storage space specially made for coding projects.

GitHub Website: <https://github.com>

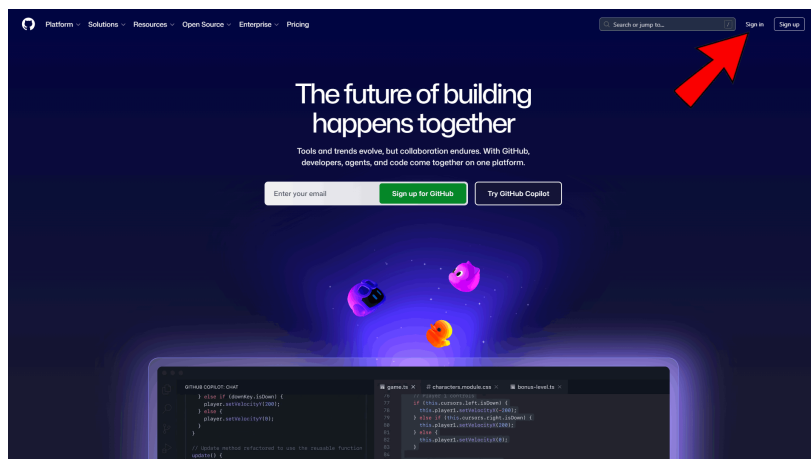
Creating a GitHub Account

Before using GitHub, we need to create a free account.

Follow the steps below carefully.

Step 1: Open the GitHub Website

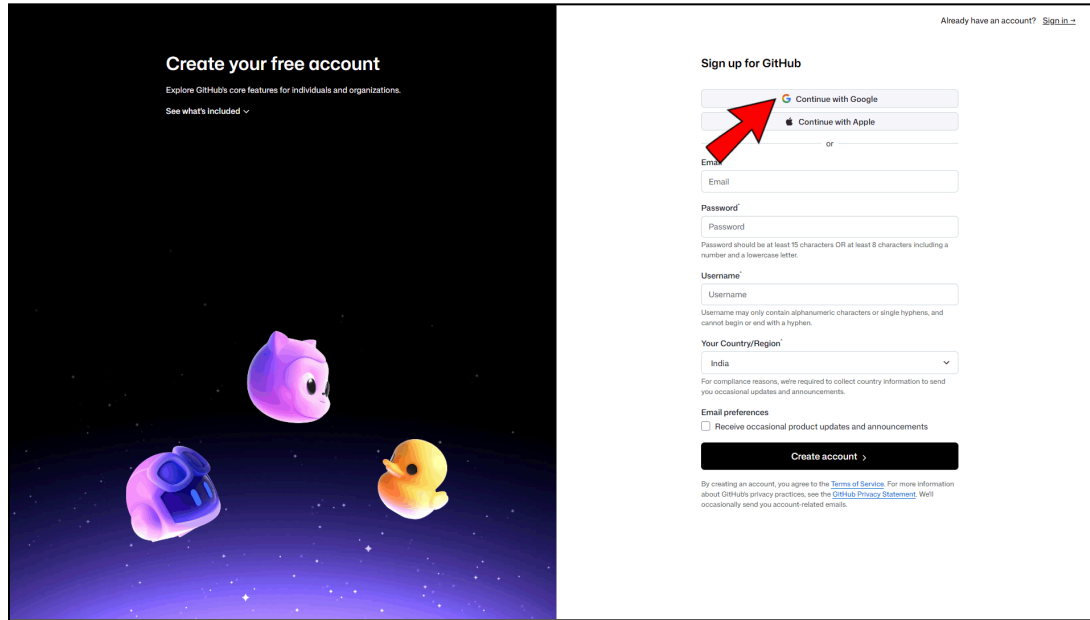
1. Open any web browser such as Chrome or Edge.
2. Click on the given link: <https://github.com>
3. When the GitHub homepage opens, click on the **Sign up** button.



Step 2: Sign Up Using a Google Account

GitHub allows users to create an account using their Google account.

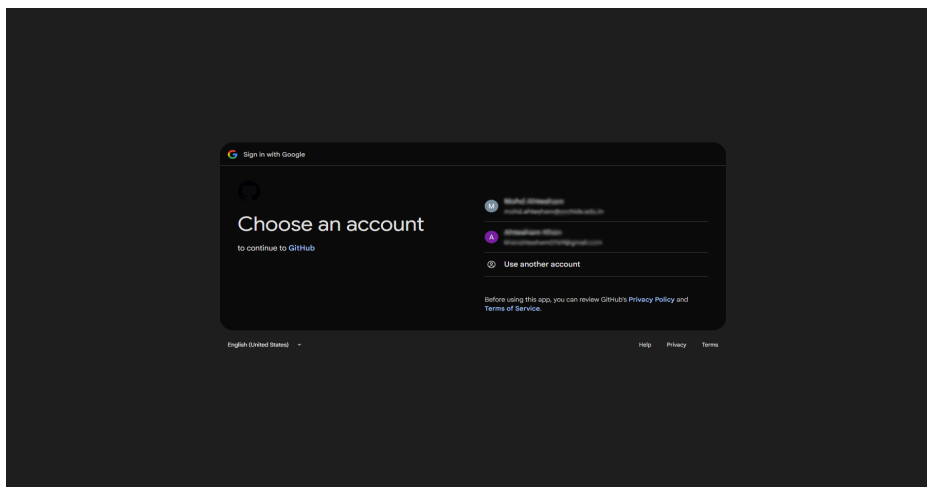
1. Click on **Continue with Google**.
2. This method is easier because your email is connected automatically.

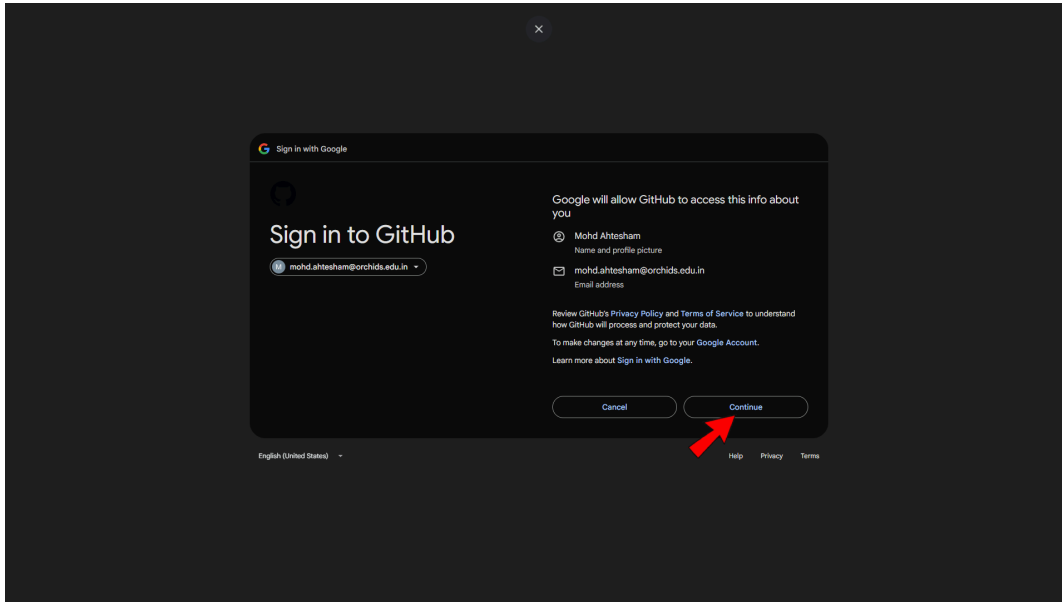


Step 3: Choose Your Google Account

After clicking the Google sign-up option, a new window will appear.

1. Select the Google account you want to use for GitHub.
2. If needed, enter your Google password.





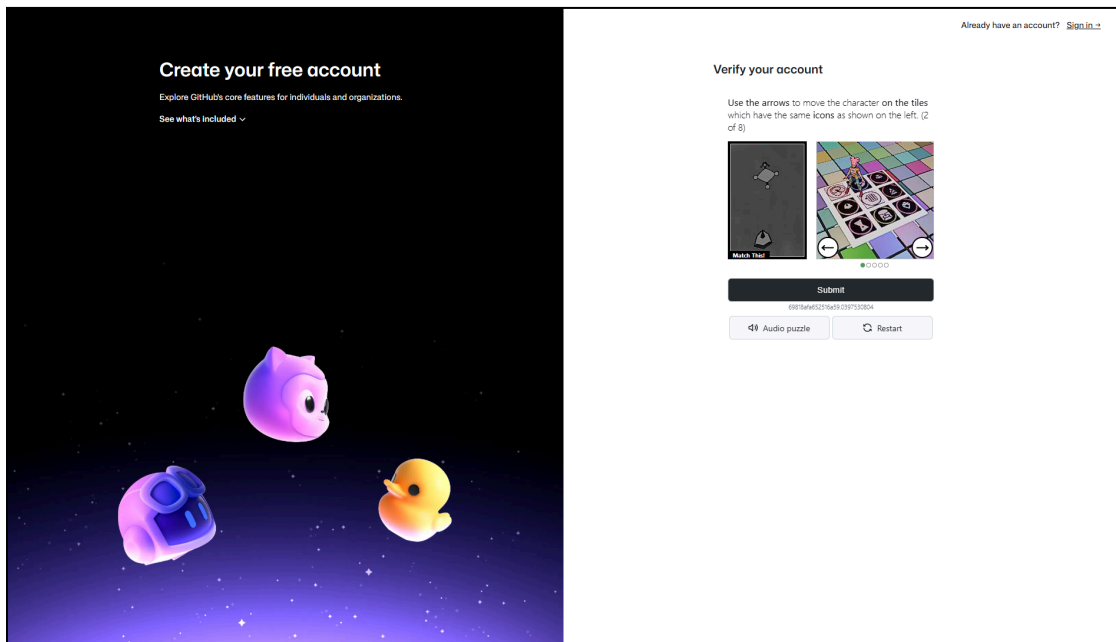
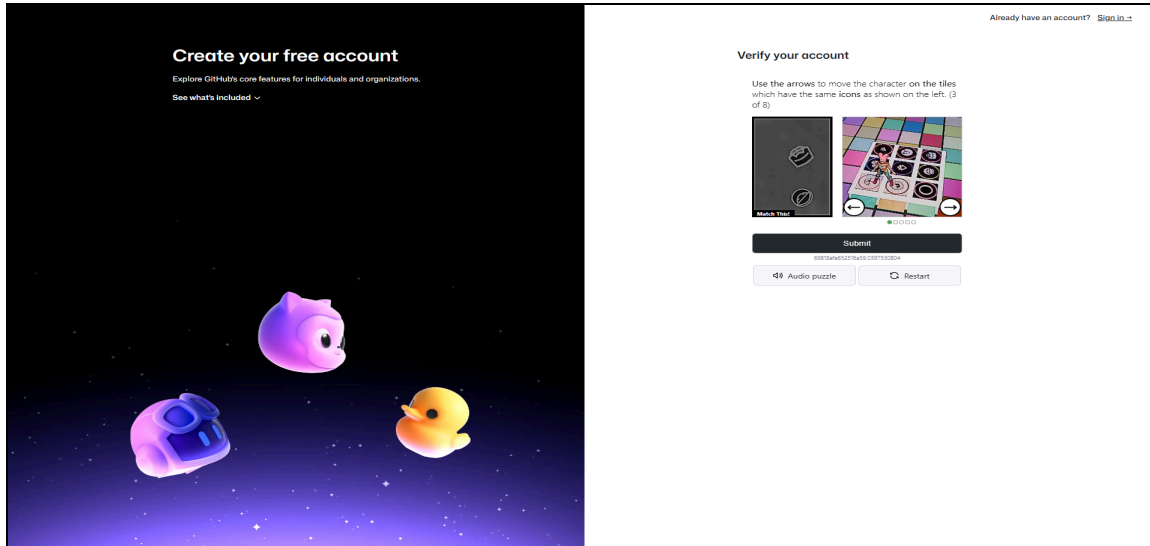
Human Verification

GitHub checks whether the user is a real person and not a robot. This process is called **Human Verification**.

Each time you may see different puzzles during this step.

Step 4: Start the Verification

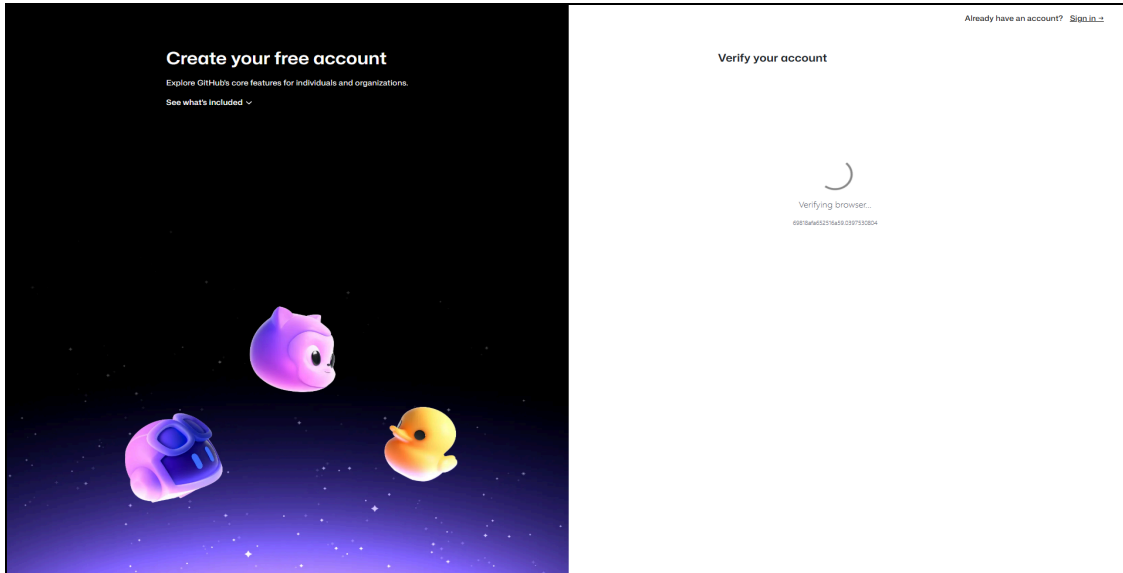
1. A verification page will appear on the screen.
2. Click on the **Visual puzzle** button to begin.
3. In the visual puzzle, a person will appear standing on the right side of the screen and two icons will appear on the left side.
4. Carefully check whether the icon under the person's feet matches the icon shown on the left side.
5. If both icons are the same, select the correct option as instructed on the screen.
6. Complete all 8 verification puzzles correctly without making mistakes to finish the verification process.



Step 5: Wait for Browser Check

GitHub may first check your browser before showing the puzzle.

1. Wait a few seconds until the checking process is complete.



Step 6: Complete the Puzzle

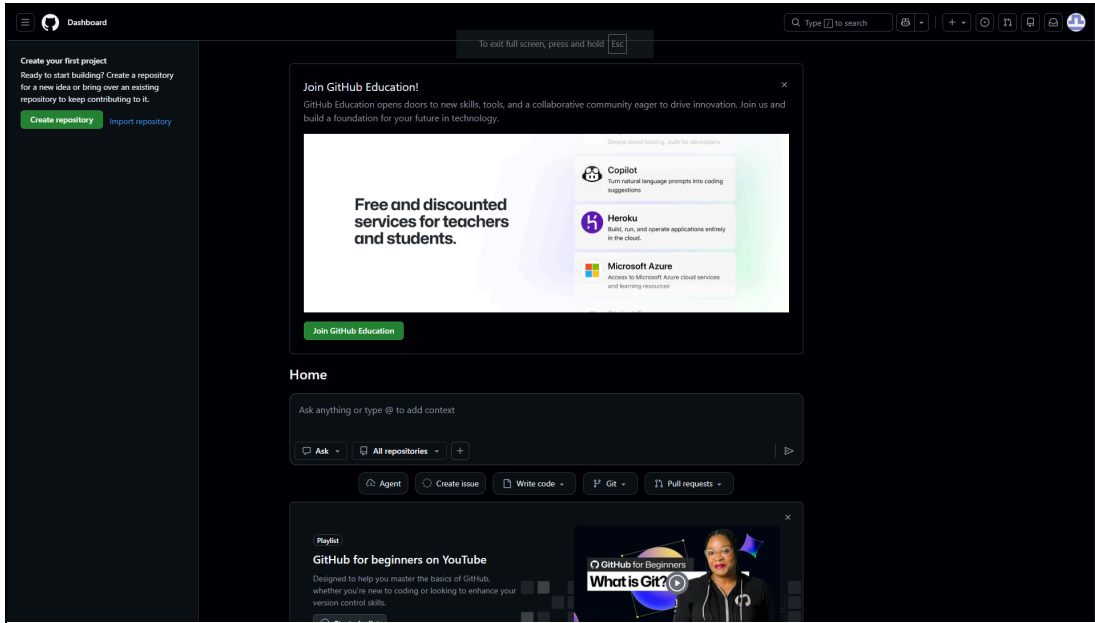
1. Carefully read the instructions shown on the screen.
2. Solve the puzzle correctly.
3. After completing the puzzle, click the **Submit** button.

Step 7: GitHub Account Successfully Created

After completing the verification, your GitHub account will be created successfully.

You can now:

- Create coding projects
- Save files online
- Explore GitHub features



Installing Git

Git is a tool that helps GitHub track project changes and upload files properly.

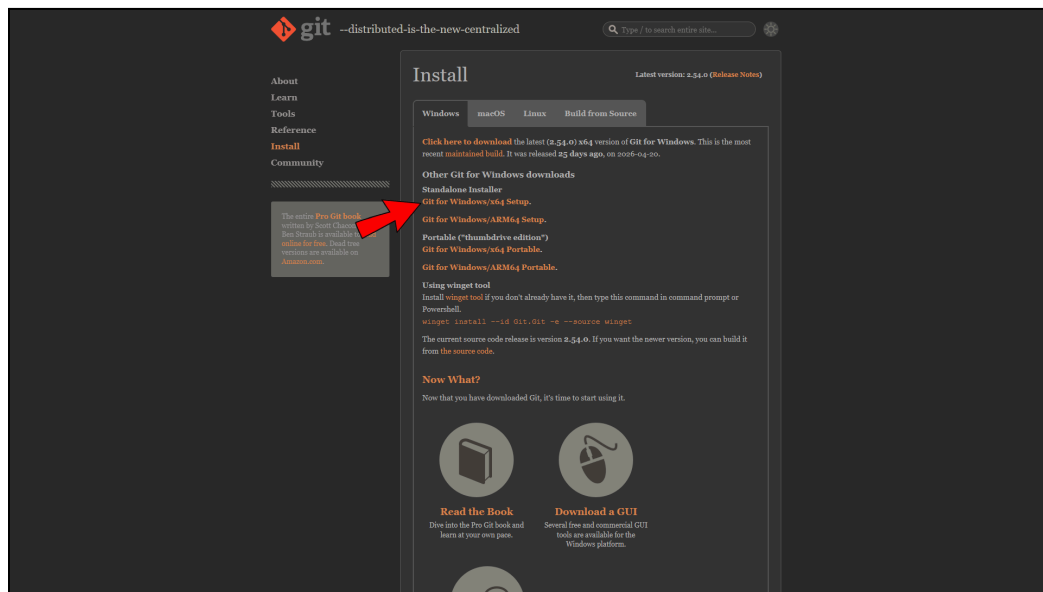
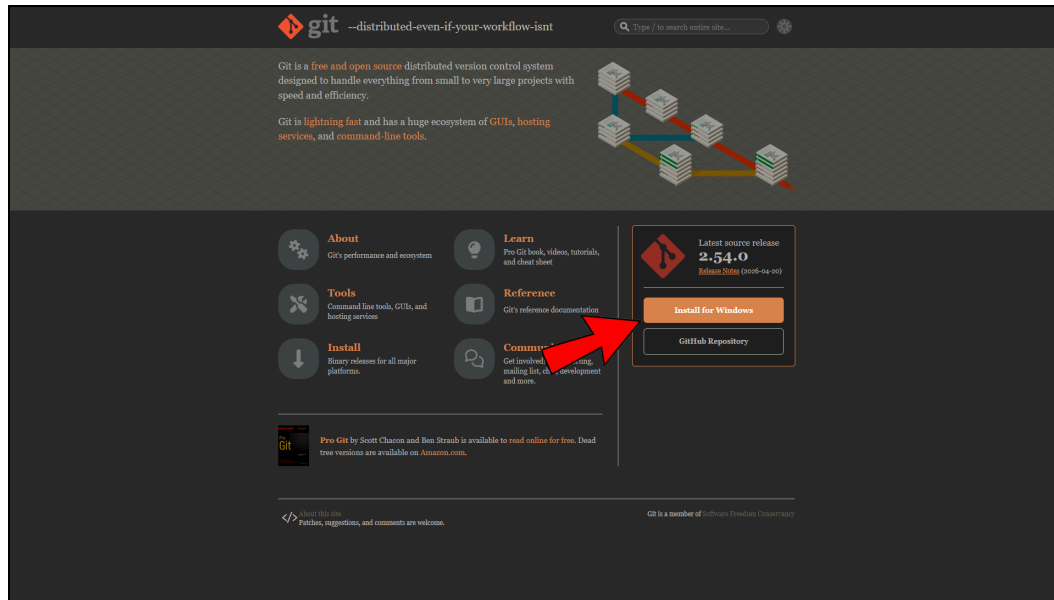
Git must be installed before using GitHub with VS Code.

Step 8: Download Git

1. Open the Git website:

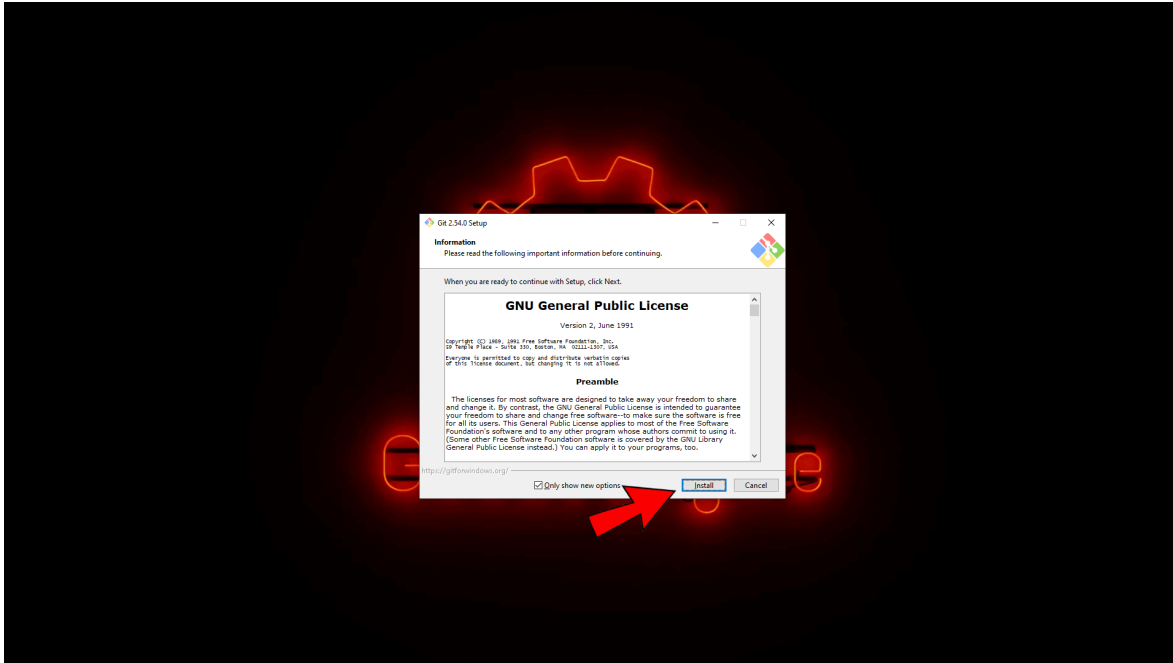
<https://git-scm.com>

2. Click on the **Download for Windows** button.



Step 9: Install Git

1. Open the downloaded Git setup file.
2. Keep all settings as default.
3. Click **Next** until the installation begins.
4. Click **Install**.



Step 10: Git Installed Successfully

After installation is complete:

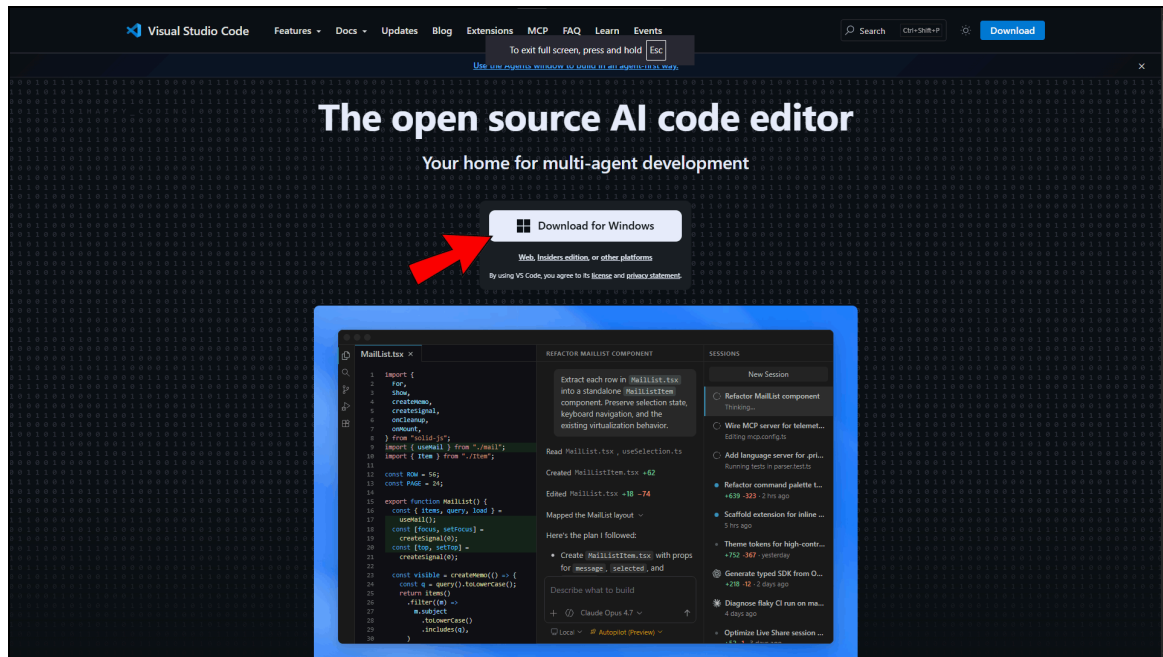
1. Click the **Finish** button.
2. Git is now ready to use with VS Code.

Installing Visual Studio Code

Visual Studio Code (VS Code) is a coding editor used to write and manage programs.

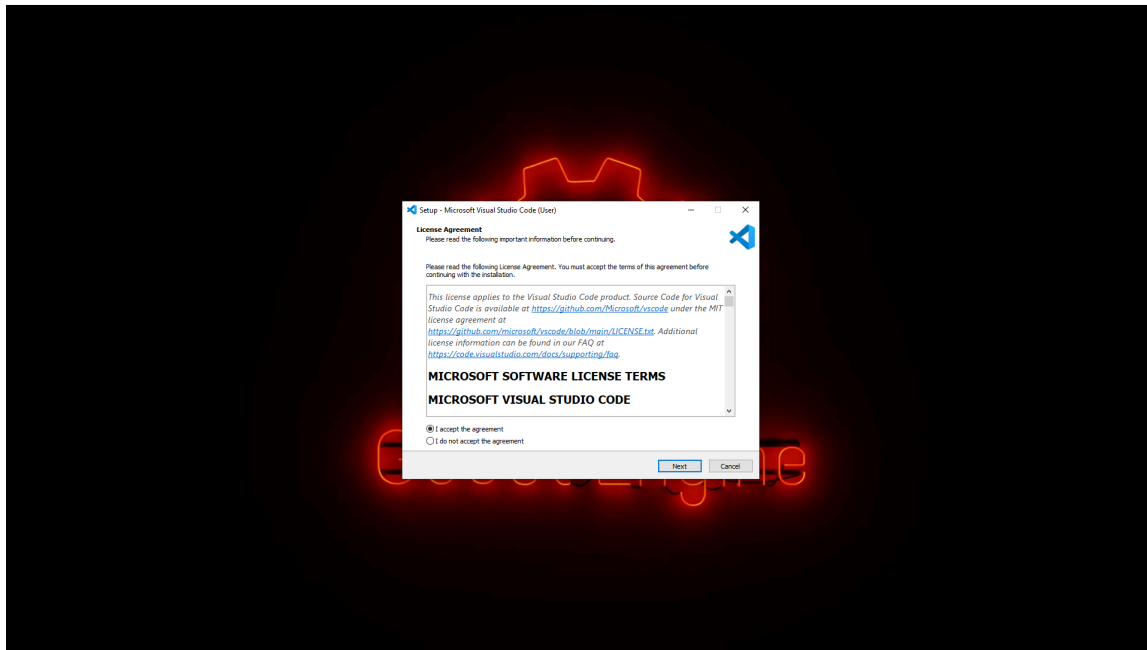
Step 11: Download VS Code

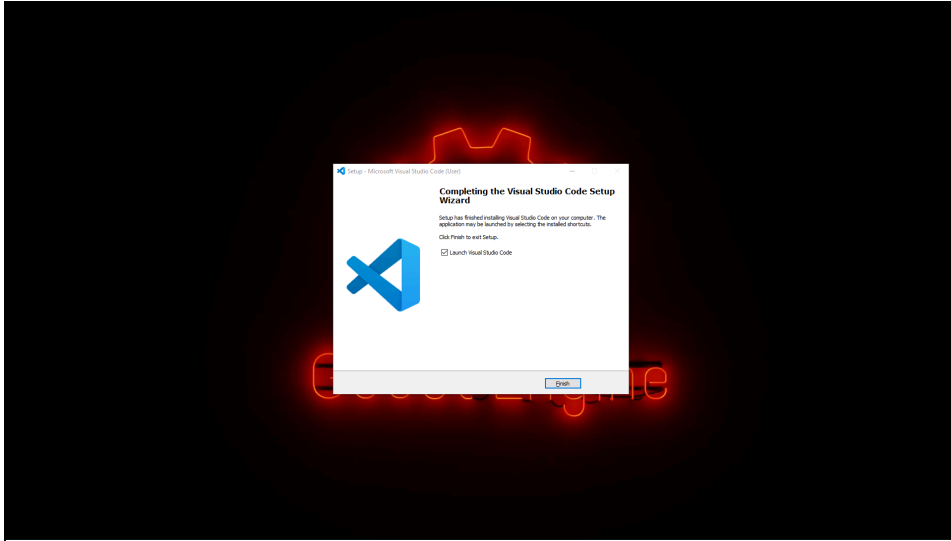
1. Open the VS Code website:
<https://code.visualstudio.com>
2. Click on the **Download for Windows** button.



Step 12: Install VS Code

1. Open the downloaded VS Code setup file.
2. Follow the installation steps.
3. Click **Finish** after installation is complete.



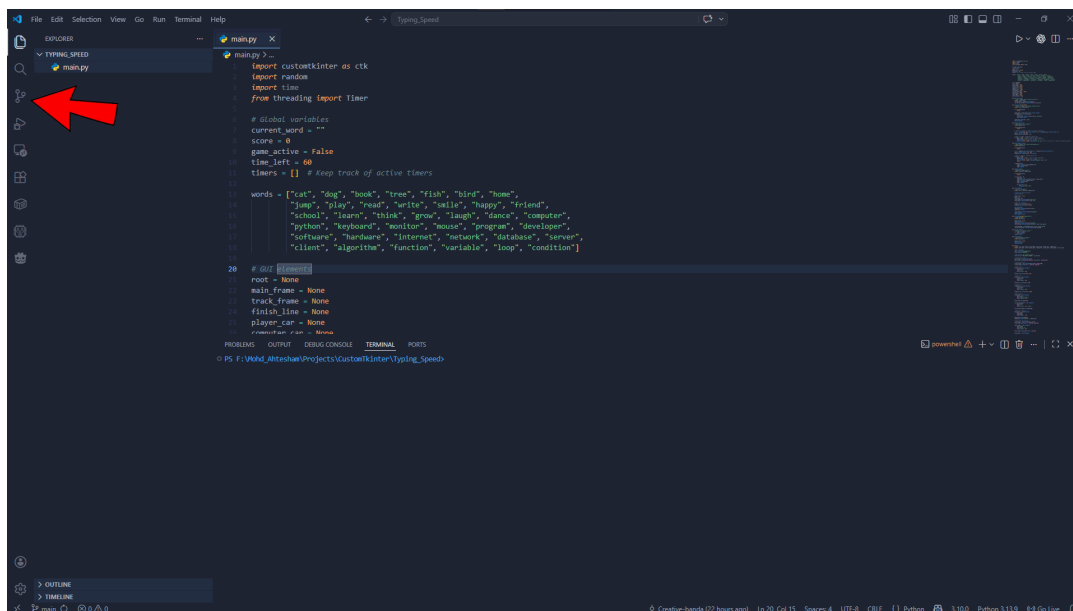


Creating a Repository from VS Code

Instead of creating repositories manually on the GitHub website, VS Code can create repositories directly.

Step 13: Open Your Project Folder

1. Open your coding project folder inside VS Code.
2. Click on the **Source Control** icon on the left sidebar.

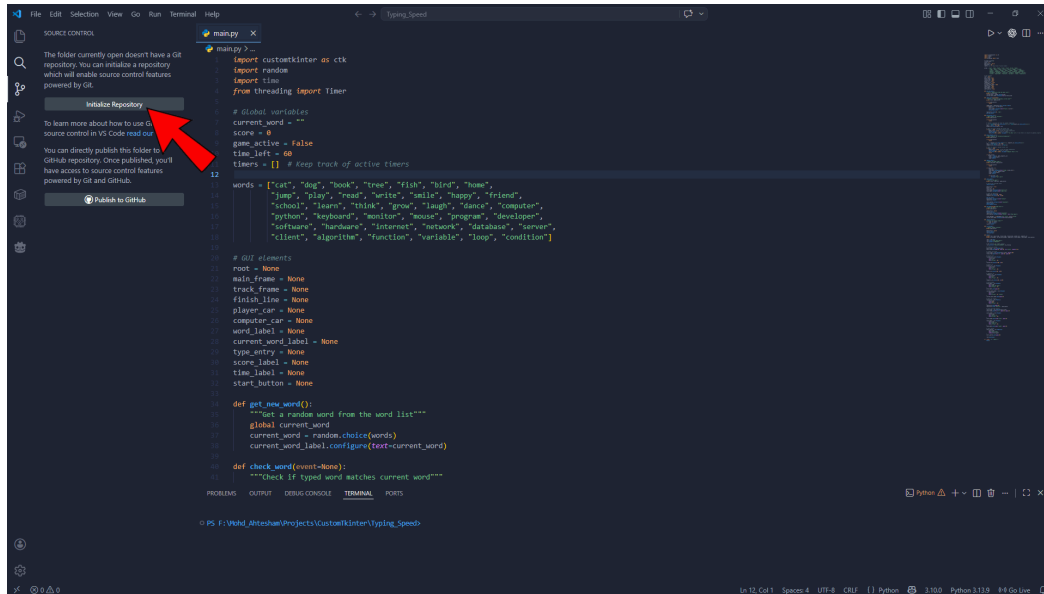


Step 14: Initialize Repository

If the project does not already have Git enabled:

1. Click on the **Initialize Repository** button.

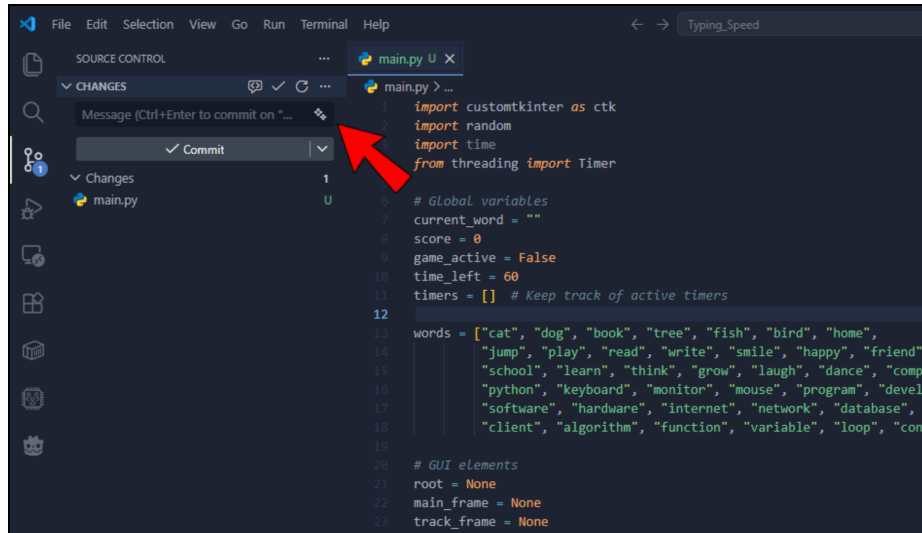
This will turn the project folder into a Git repository.



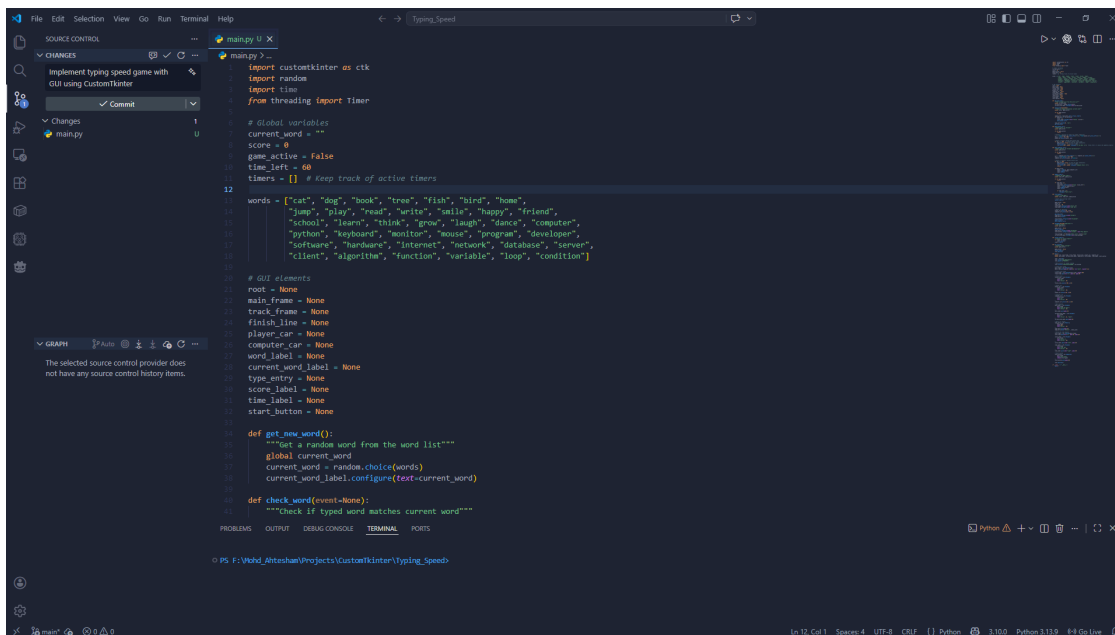
Step 15: Publish the Repository to GitHub

After initializing the repository:

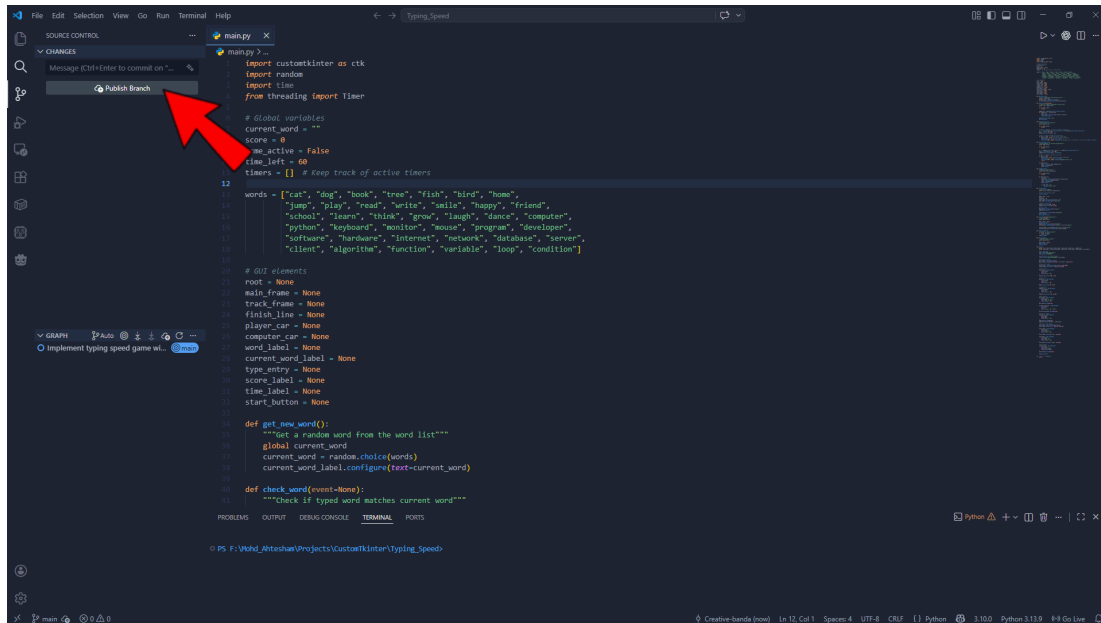
1. Make changes to any file in your project if needed.
2. Open the **Source Control** section in VS Code.
3. On the left side of the commit message box, click the magic AI icon.



4. VS Code will automatically understand the code changes and generate a commit message for you.
5. Review the generated message if needed.
6. Click on the **Commit** button to save the changes.

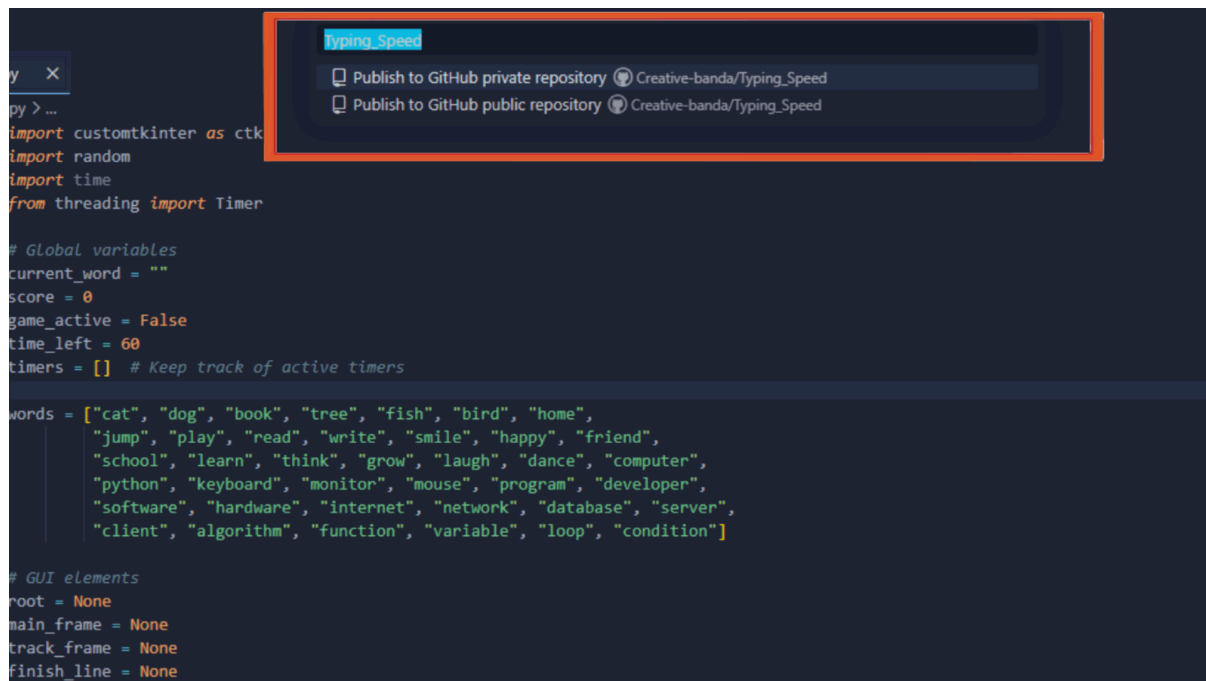


7. After the commit is completed, a **Publish Branch** button will appear.
8. Click on the **Publish Branch** button.



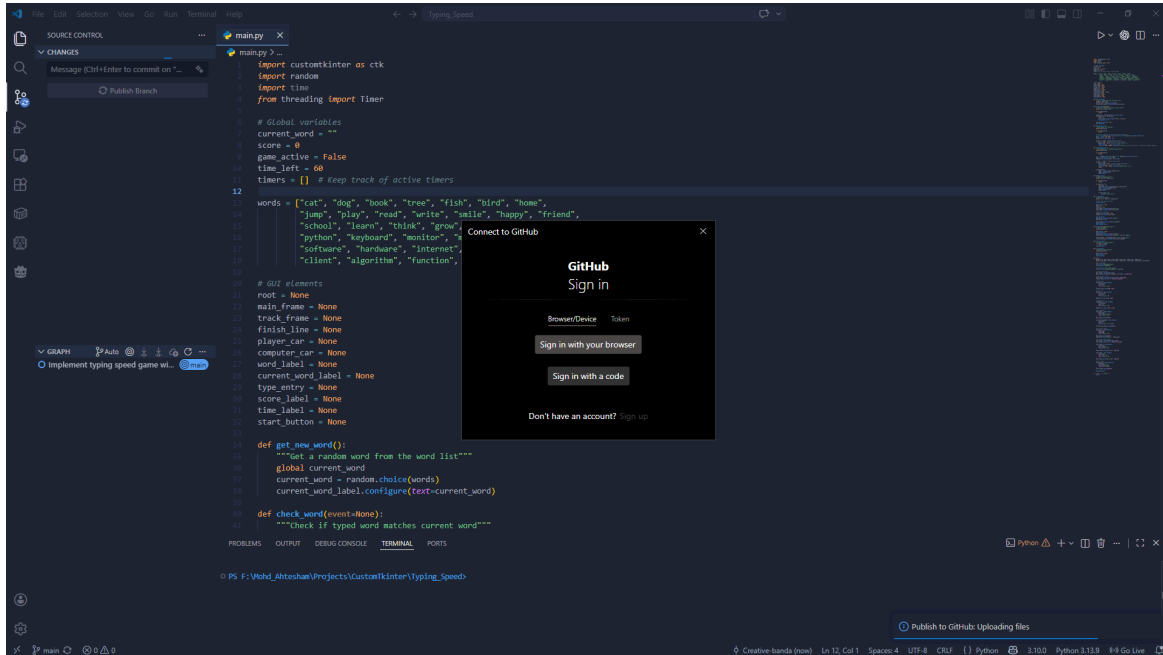
After clicking Publish Branch:

1. A selection box will appear at the top middle of VS Code asking whether the repository should be:
 - **Public** — anyone can view the repository
 - **Private** — only you can access the repository
2. Select either **Public** or **Private** based on your choice.



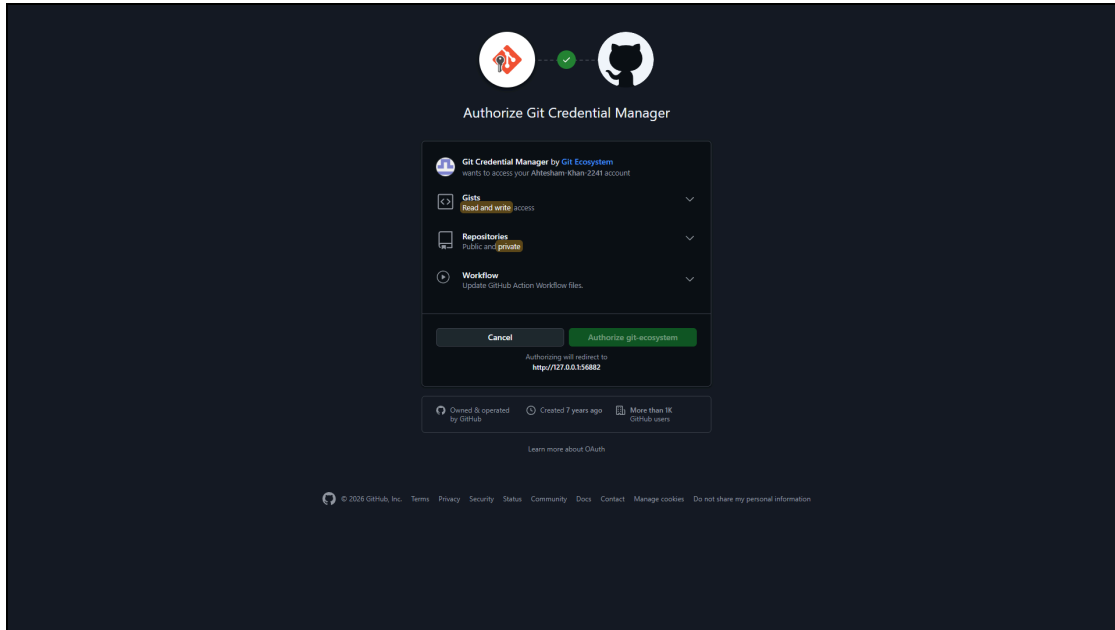
After selecting the repository type:

1. A GitHub sign-in popup window will appear.
2. Select **Sign in with your browser**.



Your web browser will now open automatically.

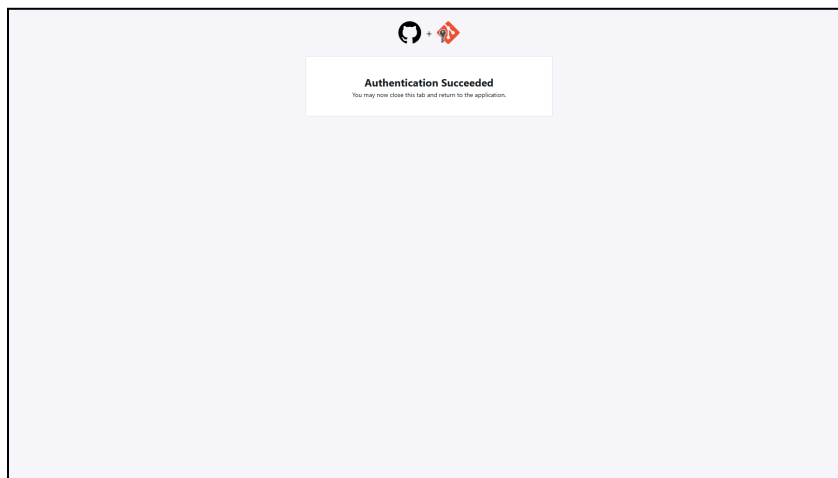
1. Click on **Continue with Google**.
2. Select your Google account.
3. Click on **Authorize git-ecosystem** when permission is requested.



After successful login:

1. A message saying **Authentication Succeeded** will appear in the browser.
2. Return to VS Code.

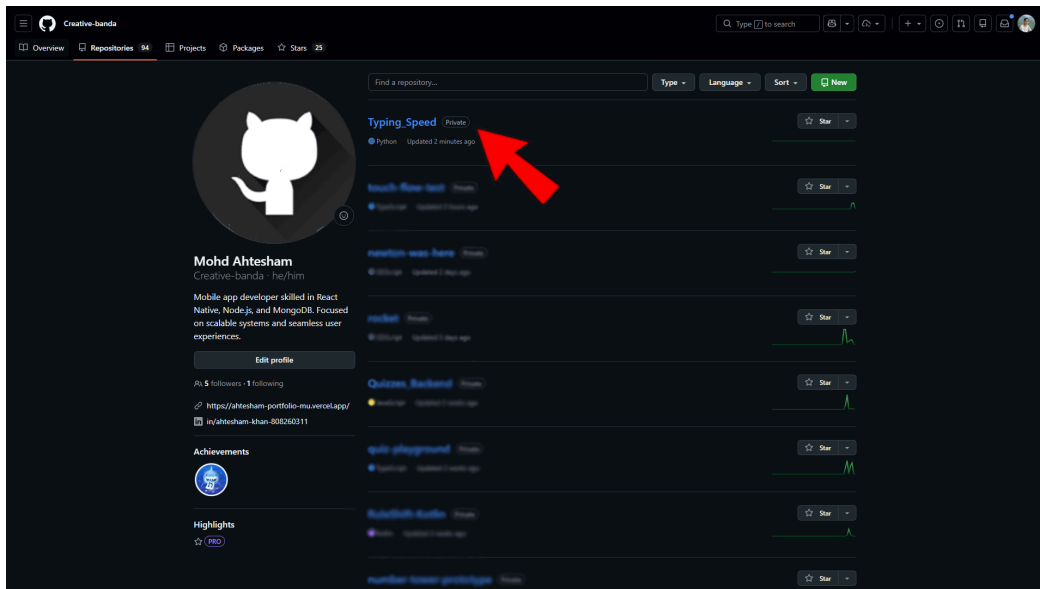
Your repository will now be published successfully to GitHub.



Step 16: Repository Published Successfully

After clicking the **Publish to GitHub** button, VS Code will upload the project to GitHub automatically.

1. Wait for the upload process to finish.
2. Your repository will now be available online.



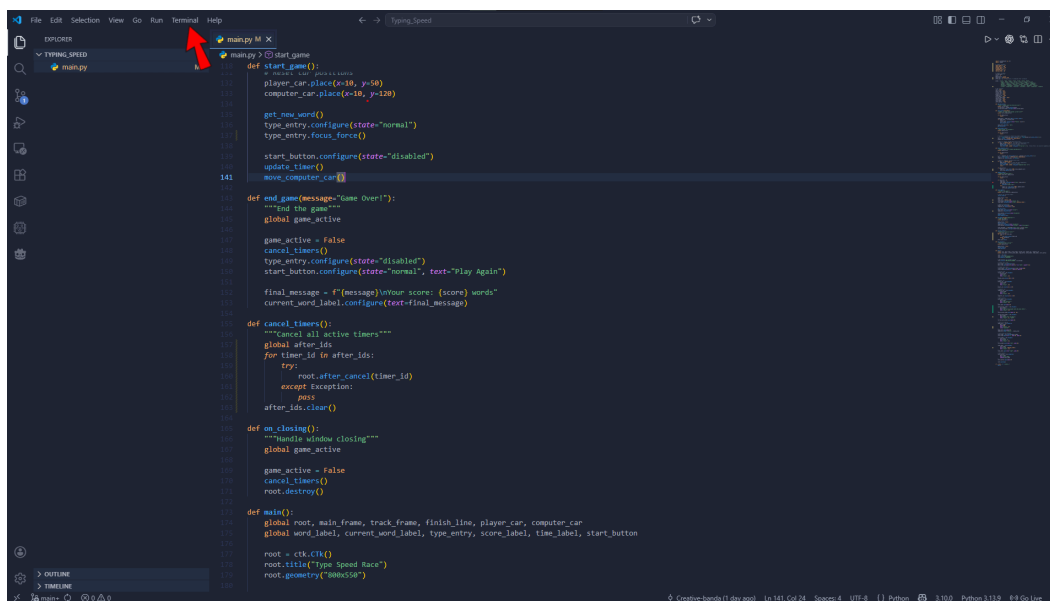
Configuring Git User Information

Before saving commits, Git needs your name and email address.

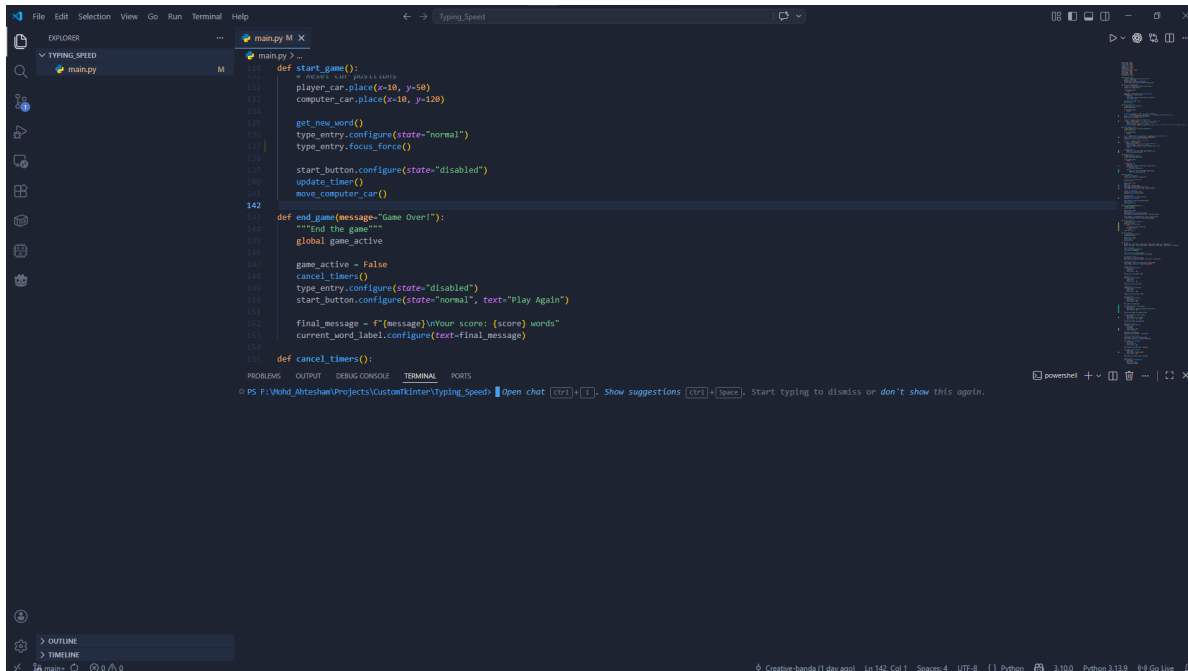
This setup is required only one time on a computer.

Step 17: Open the Terminal in VS Code

1. Open VS Code.
2. Click on the **Terminal** menu at the top.
3. Select **New Terminal**.



The terminal will open at the bottom of VS Code.



Step 18: Set Your Git Username

Type the following command inside the terminal:

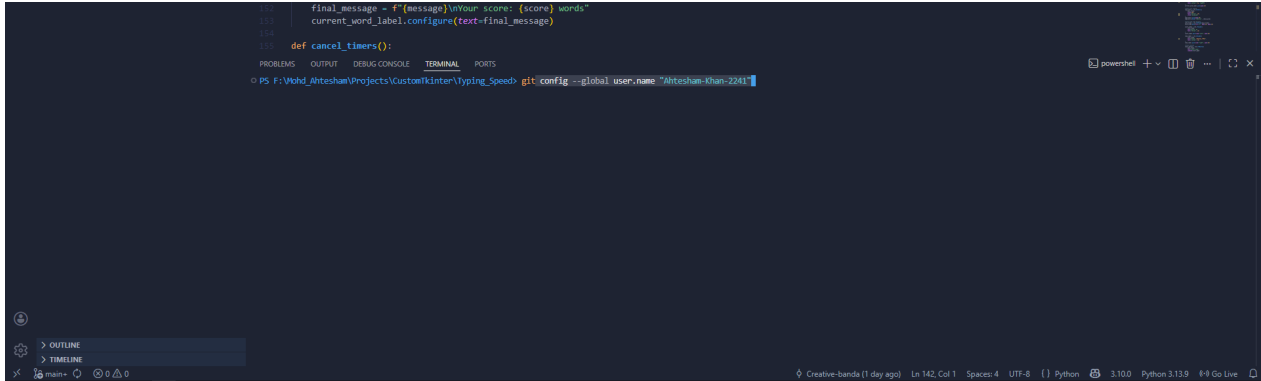
```
git config --global user.name "Your GitHub Name"
```

Replace "Your GitHub Name" with your real name or GitHub username. You can find that name in your github profile

Example:

```
git config --global user.name "Harsh"
```

Press **Enter** after typing the command.



The screenshot shows a VS Code editor with a Python script in the background. The script contains the following code:

```
133 final_message = f'{message}\nYour score: {score} words'
134 current_word_label.configure(text=final_message)
135
136 def cancel_timers():
137     pass
```

The terminal window in the foreground shows the command `git config --global user.name "Ahtesham-khan-2241"` being executed. The terminal output shows the command being run and the user name being set.

Step 19: Set Your Git Email Address

Now type the following command:

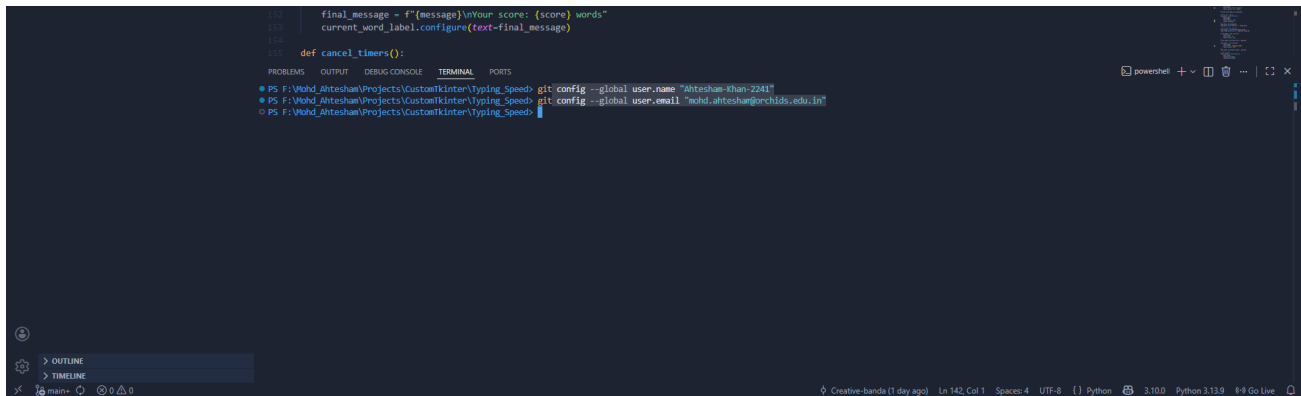
```
git config --global user.email "youremail@example.com"
```

Replace the email with your GitHub email address.

Example:

```
git config --global user.email "example@gmail.com"
```

Press **Enter** after typing the command.



The screenshot shows a VS Code editor with a Python script in the background. The script contains the following code:

```
133 final_message = f'{message}\nYour score: {score} words'
134 current_word_label.configure(text=final_message)
135
136 def cancel_timers():
137     pass
```

The terminal window in the foreground shows the command `git config --global user.email "mohd.ahmedgarcia@edu.in"` being executed. The terminal output shows the command being run and the email address being set.

Step 20: Why This Step is Important

Git uses this information to identify who made the project changes.

Without setting the username and email, Git may show an error while committing changes.

After completing this setup, you can successfully:

- Commit changes

- Publish repositories
- Sync projects with GitHub

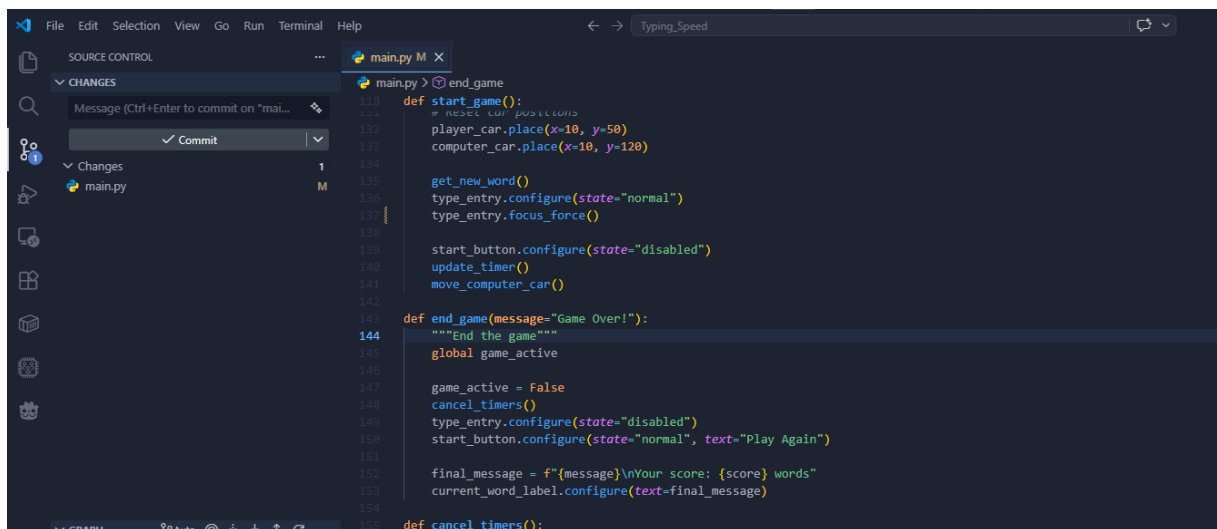
Making Changes in the Project

GitHub can track all changes made inside a project.

Step 21: Edit or Add Files

1. Open any file in your project.
2. Make some changes or create a new file.
3. Save the file.

VS Code will automatically detect the changes.



Committing Changes

A commit is like saving your work with a short message.

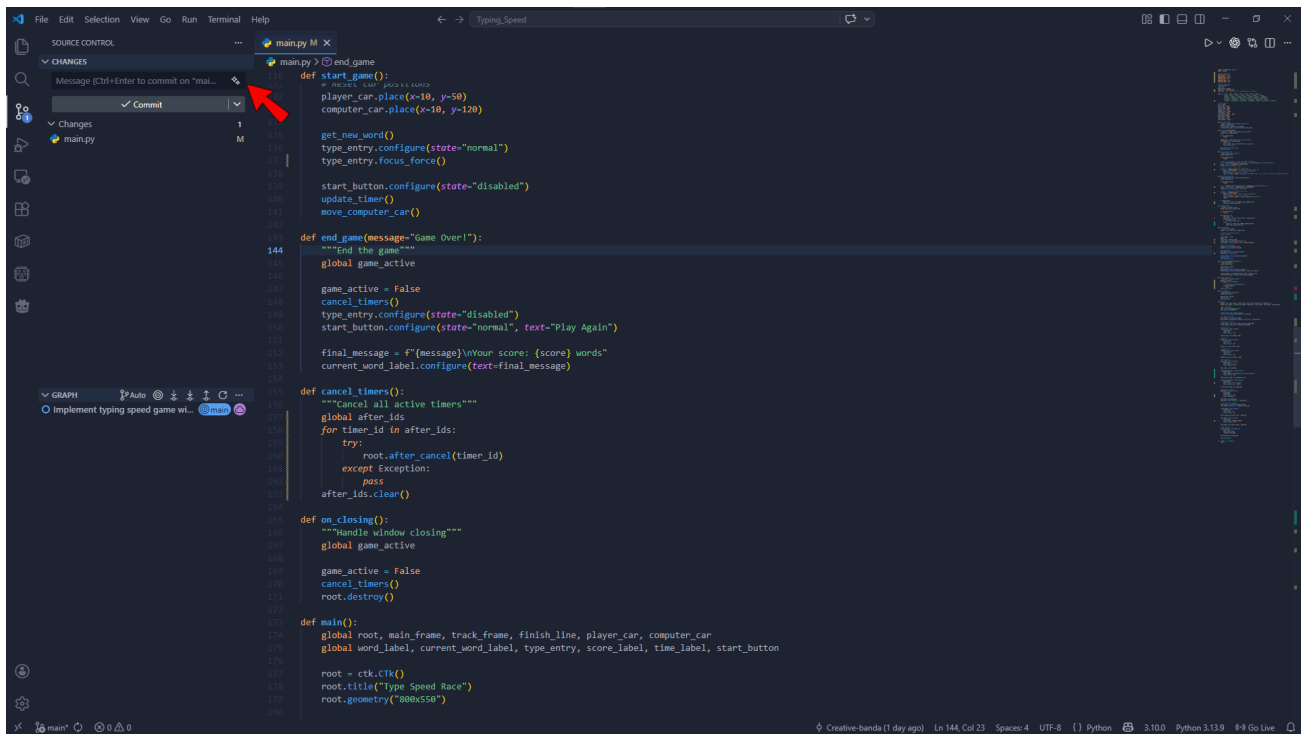
Step 22: Write a Commit Message

- Open the **Source Control** section.
- Click on the **Magic AI** icon near the commit message box.

- VS Code will automatically generate a commit message based on the changes made in the project.
- Review the generated message if needed.

Example generated messages:

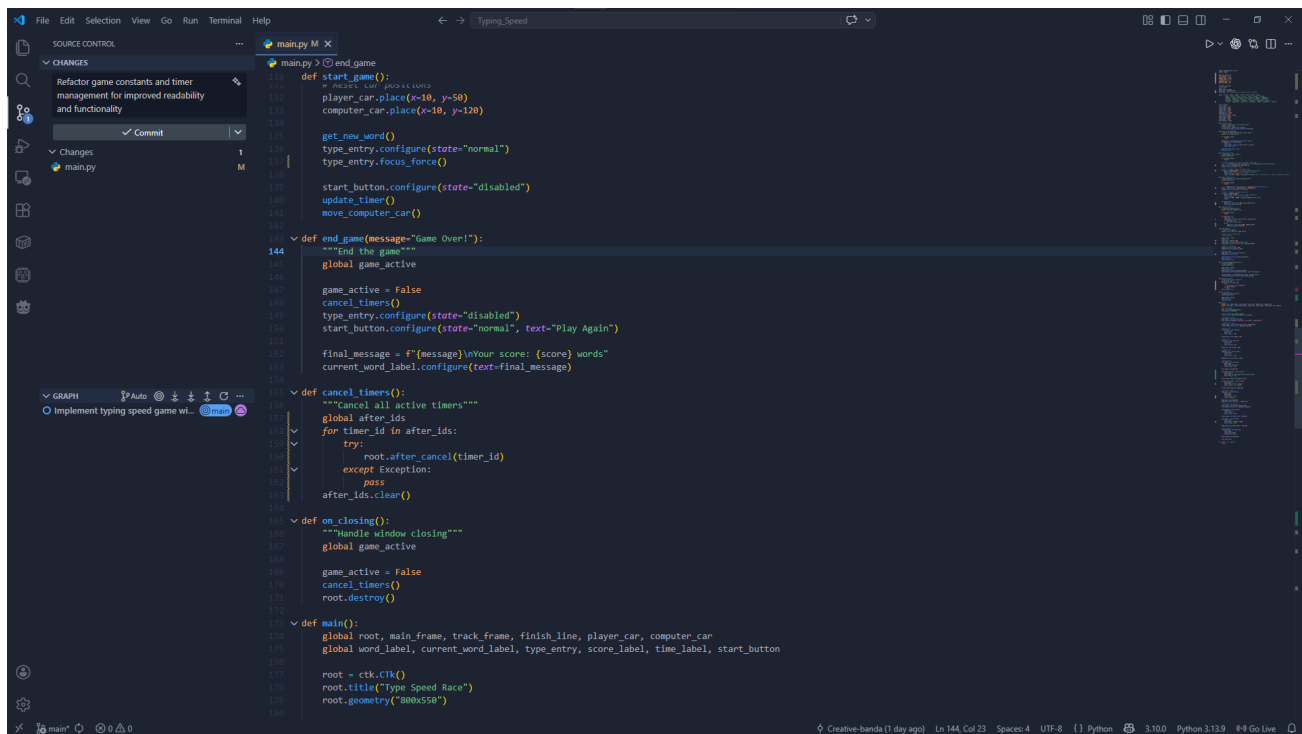
- Added new feature
- Updated project
- Fixed typing mistake



Step 23: Commit the Changes

1. Click on the **Commit** button.

The changes are now saved inside Git.



Syncing Changes to GitHub

Syncing uploads the latest project changes to GitHub.

Step 24: Sync Changes

1. Click on the **Sync Changes** button in VS Code.
2. Wait for the upload process to complete.

[Insert screenshot of Sync Changes button]

Step 25: View Updated Repository on GitHub

1. Open your repository on GitHub.
2. The latest project changes will now appear online.

[Insert screenshot of updated repository on GitHub]

Conclusion

GitHub and VS Code make it easy to manage coding projects.

In this tutorial, you learned how to:

- Create a GitHub account
- Complete human verification
- Install Git
- Install VS Code
- Install the GitHub extension
- Initialize a repository
- Publish a project to GitHub
- Commit project changes
- Sync changes to GitHub

These tools are widely used by programmers and students around the world for coding and project management.